

# TFPT — $E_8$ Audit, Cascade Bridge and Bootstrap

The seven  $E_8$  slices as an audit raster, the old cascade as the same even-integer spine, and the Möbius self-consistency loop

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## What this document is about

$E_8$  as an *audit container*, not a mystery: the seven maximal slices of 248 as a falsification raster (every load-bearing number must appear in at least one projection), the bridge showing the old  $E_8$  orbit cascade  $D=60-2n$  is the same even-integer spine as the compiler, and the Möbius bootstrap in which  $g_{\text{car}} = 5$  and the “8” in  $c_3$  are overdetermined  $E_8$ -closure fixed points (only  $\pi$  irreducible).

**What  $E_8$  is *not* (and why the known no-go results do not apply)**

$E_8$  here is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures — **not** an unbroken physical gauge group. The Standard Model is a *readout* after projection (not a direct  $E_8$  gauge theory), the Lorentz signature appears only after projection, and fermions are not “everything in the adjoint”. Consequently the standard objections to literal  $E_8$  world-formulas do *not* bite: Distler–Garibaldi (no chiral fermions / representation obstructions for  $E_8$  as a 4d gauge group) and Coleman–Mandula (no nontrivial mixing of spacetime and internal symmetry) both constrain  $E_8$  as a *spacetime gauge symmetry*, which TFPT does not claim. The defensible statement is: *TFPT is the unique parabolic compilation of the anchor  $a = (1, 1, 2)$  into the  $E_8$  audit hull* — a discrete classifier, attacked on its own (combinatorial) terms, not on gauge-theoretic ones.

**The TFPT document set — what is treated where**

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R. tfpt\_research\_contracts** — the open interfaces ( $v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$ ) as numbered contracts; **H. tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O. origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

**You are reading document 3 ( $E_8$  Audit & Bootstrap).**

**TFPT status at a glance — read this card the same way in every document**

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

| Interface             | Question                                                                                        | Status        |
|-----------------------|-------------------------------------------------------------------------------------------------|---------------|
| $v_{\text{geo}}$      | the one metrology unit ( $= 1/\sqrt{G} = m/\mu$ ); No-Unit Thm: no compiler scale               | primitive [O] |
| $G_{\text{net}}$      | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ | [C]           |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$ , axion, $m_p/m_e$ )                        | [C]           |

**Gravity is parameter-free.** The classical metric-sector field equation is no longer only an  $R+R^2$  readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  with *both* coefficients fixed —  $c_3^{-1} = 8\pi$  (no free dimensionless Newton dial; the thermodynamic origin  $2\pi/\eta$  *coincides* with the geometric one  $|\mathbb{Z}_2|2\pi\chi$  via  $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$ , so  $c_3$  is triply over-determined — anchor, geometry, thermodynamics, v358) and  $\Lambda$  from  $\alpha$  ( $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ , v60); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (v359). The residual here is only the equation-of-state status and the one unit  $v_{\text{geo}}$ ; the global measure (C7 = QG.AMB.01) is now a [C] redundancy (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann.

The seam-net structural theorem  $G_{\text{net}}$  (= SEAM.EQUIV.01: the raw seam *is* the holomorphic  $(E_8)_1$  net) is now *closed modulo a cited theorem*. The explicit lattice model (v367/v368) and the  $S_3$  closure stack pin the target at every computable level — central charge  $c = 8$  (v376), the  $(E_8)_1$  character with 248 currents and one primary (v377), genus-one torus  $\text{GSD} = 1$  (v378) and reflection positivity (v379) — and it is Lean-formalised as FORM.SEAM.MMST.01: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the #print axioms check is clean. The *only* residual that stays [O] is the abstract continuum existence of the scaling limit, i.e. exactly those two cited published theorems (v336); so this is *closed modulo a cited theorem*, not solved. QGEO.SYM.01 (the  $\mu_4$  deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (v335, Lean FORM.QGEO.BW.01). The nonperturbative quantum-gravity measure QG.AMB.01 is no longer carried as an open item: it is a [C] *redundancy* (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin  $1.648 > 0$ , v76/v330). So the honest residual is the one unit  $v_{\text{geo}}$  plus the typed  $F_{\text{transfer}}$  interfaces, above the cited-theorem ceiling on the seam and with QG.AMB.01 a [C] redundancy. Everything carries a claim ID resolving to a row of *verification/status\_ledger.csv* (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ( $U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus F_{\text{transfer}}$ , with the seam  $G_{\text{net}}$  closed modulo a cited theorem and QG.AMB.01 a [C] redundancy.

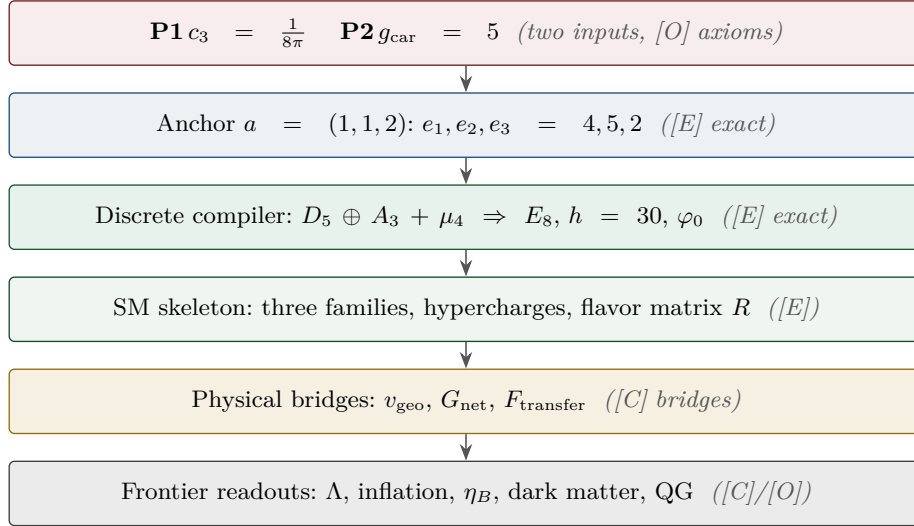
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[E] exact / machine-proven

[C] conditional

[O] open / axiom

[X] kill test



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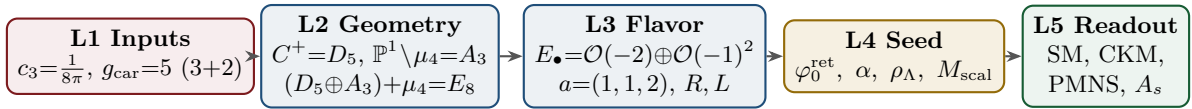
## Part I

# The $E_8$ secondary-branching atlas

**Purpose and discipline.** This note does *not* introduce new physics. It builds an *audit raster*: the claim that *every load-bearing TFPT number should appear in at least one  $E_8$  branching projection*. A number that appears in none is suspect; a number appearing in several independent projections becomes structurally strong. All arithmetic identities below are exact [E]; the *physical readings* are audit-level [O] unless tied to a derivation. (All 77 numeric claims were machine-checked with exact integer/rational arithmetic.)

## 1 The compiler core (where this note sits)

The strongest current form of the theory is a five-layer compiler:



$E_8$  is now a *closure* object, not an input: the carrier is the  $D_5 = \mathfrak{so}_{10}$  half-spinor, the four punctures  $\mathbb{P}^1 \setminus \mu_4$  give  $A_3 = \mathfrak{su}_4$ , and the common discriminant group  $\mathbb{Z}_4$  glues them, with the norm test  $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2$ . This note charts the *secondary* maximal subalgebras of  $E_8$  as audit windows onto the TFPT modules.

## 2 The $(1, 1, 2)$ anchor microcode

The parabolic anchor  $a = (1, 1, 2)$  (splitting type of  $E_\bullet = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ ) is a miniature compiler. Its power sums  $p_n(a) = 1^n + 1^n + 2^n = 2 + 2^n$  generate the discrete budget:

| $n$ | $p_n = 2 + 2^n$ | TFPT reading (verified)                                                           |
|-----|-----------------|-----------------------------------------------------------------------------------|
| 1   | 4               | $ \mu_4  = -\deg E_\bullet$ (glue index)                                          |
| 2   | 6               | $ R^+(A_3) $ (positive $A_3$ roots) = $\mathbb{Z}_6$ cusp order                   |
| 3   | 10              | $\mathcal{A}_\Lambda = \binom{5}{2} = \dim V_{D_5}$ (action ladder / pair sector) |
| 4   | 18              | $N_{\text{fam}}  R^+(A_3)  = 3 \cdot 6$                                           |
| 5   | 34              | $2h(D_5) + 3 R^+(A_3)  = 16 + 18$ ; and $34 -  R(A_3)  = 22 = \sum R$             |

and the budget compresses to products of power sums:

$$q(A_3) = \frac{p_2}{2p_1} = \frac{3}{4}, \quad q(D_5) = \frac{p_3}{2p_1} = \frac{5}{4}, \quad \sum L = p_1 p_3 = 40,$$

$$\Omega_{\text{adm}} = 2p_1 p_2 = 48, \quad D_{\text{start}} = p_2 p_3 = 60, \quad |R(E_8)| = 4p_2 p_3 = 240.$$

The first three power sums alone reach  $E_8$ :  $p_1 p_2 p_3 = 4 \cdot 6 \cdot 10 = 240 = |R(E_8)|$ , and  $p_1 p_2 p_3 + (p_4 - p_3) = 240 + 8 = 248 = \dim E_8$ .

### The anchor difference ladder: the binary carrier spine [E]

While the *products* of the power sums give the large budgets, the *differences* give the binary carrier spine in one line:

$$p_n(a) = 2 + 2^n \implies \Delta p_n = p_{n+1} - p_n = 2^n : \quad (2, 4, 8, 16) = (|\mathbb{Z}_2|, |\mu_4|, h(D_5), \dim S^+).$$

So the single anchor  $a = (1, 1, 2)$  generates the big numbers multiplicatively and the doubling spine 2, 4, 8, 16 additively — one vector, two readings.

### The anchor reconstructs the carrier rank — and the 41-chain

The characteristic polynomial of the anchor vector is  $(t-1)^2(t-2) = t^3 - 4t^2 + 5t - 2$ , with elementary symmetric data

$$e_1(a) = 4 = |\mu_4|, \quad e_2(a) = 5 = g_{\text{car}}, \quad e_3(a) = 2 = |\mathbb{Z}_2|_{\text{sheet}}.$$

So the parabolic flavor geometry reads the carrier rank  $g_{\text{car}} = 5$  *backwards*:  $D_5 \Rightarrow a$  but also  $a \Rightarrow g_{\text{car}}$ . Moreover  $3^2 + 4^2 = 5^2$  sits as  $(p_0, e_1, e_2) = (3, 4, 5)$ , giving the *anchor form* of the abelian index

$$10b_1 = e_1(a)^2 + e_2(a)^2 = 4^2 + 5^2 = 41$$

alongside the known  $41 = 40 + 1 = \sum L + N_{\Phi}$ . [E]

### Pascal closure: $g_{\text{car}} = 5$ is the unique solution (kills alternatives) [E]

The carrier rank is not merely a good fit — it is the *unique* root of an exact closure. The even-exterior carrier dimension  $2^{g-1}$  must equal the truncated Pascal sum  $\binom{g}{0} + \binom{g}{1} + \binom{g}{2}$  (the  $1+g+\binom{g}{2}$  that builds  $\dim S^+$ ):

$$2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2} \iff g = 5$$

( $g=3 : 4 \neq 7$ ;  $g=4 : 8 \neq 11$ ;  **$g=5 : 16=16$** ;  $g=6 : 32 \neq 22$ ;  $g=7 : 64 \neq 29$ ). At  $g = 5$  the row is  $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$ , the same Pascal triple that reads family, action ladder and the  $E_8$  root count  $16 \cdot 5 \cdot 3 = 240$ . This is the fourth independent lock on  $g_{\text{car}} = 5$  (with rank-fill, Coxeter-match, and integer-glue/norm), and unlike those it explicitly *kills* the neighbours  $g = 4, 6$ .

**The Pascal Ladder Theorem** ([[verification/v108\\_pascal\\_ladder.py](#)]). The closure generalises exactly: for *every* truncation degree  $K$ ,

$$2^{g-1} = \sum_{k \leq K} \binom{g}{k} \iff g = 2K + 1$$

(odd  $g$  solves by the binomial midpoint identity; even  $g$  is excluded by the strict straddle  $\sum_{k < g/2} < 2^{g-1} < \sum_{k \leq g/2}$ ). **Consequence (zero slack): the carrier selection  $g = 5$  is *equivalent* to the truncation degree  $K = 2$**  — the Pascal-*selection* residual of the red team (Target B) is therefore exactly the one named hypothesis *Quadratic Boundary Locality* (the seam certifies code degrees  $\leq 2$ ; the architecture document). Neighbour worlds on the ladder:  $K=1 \rightarrow g=3$  (a *one-family* world),  $K=2 \rightarrow g=5$  (three families),  $K=3 \rightarrow g=7$  (*nine* families),  $K=4 \rightarrow g=9$  *inconsistent* ( $N_{\text{fam}} = 255/9 \notin \mathbb{Z}$ ). Overdetermination: among the ladder worlds, only  $K = 2$  also satisfies the independent rank closure  $g + N_{\text{fam}} = 8 = \text{rank } E_8$  — two independent selections agree on the same world. The hypothesis itself stays [C].

### The seed is an anchor expression

The retained seed is exactly the two-term anchor combination of  $c_3$ :

$$\varphi_0^{\text{ret}} = \frac{2p_1}{p_2} c_3 + 2p_1 p_2 c_3^4 = \frac{4}{3} c_3 + 48 c_3^4 = \frac{1}{6\pi} + \frac{3}{256\pi^4}$$

(verified to 30 digits). The Cabibbo base is  $\lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})}$ , the hierarchy engine of the whole fermion spectrum. [E]

## 3 The residue and length matrices as spectral objects

With the residue matrix  $R$  and the full word-length matrix  $L = R + 6W$  ( $W$  the rank-one winding),

$$R = \begin{pmatrix} 1 & 3 & 0 \\ 1 & 5 & 2 \\ 2 & 5 & 3 \end{pmatrix}, \quad L = \begin{pmatrix} 7 & 3 & 0 \\ 7 & 5 & 2 \\ 8 & 5 & 3 \end{pmatrix}, \quad W = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

the spectral invariants are pure compiler numbers (all verified):

| Invariant of $R$      | value                  | Invariant of $L$      | value                                  |
|-----------------------|------------------------|-----------------------|----------------------------------------|
| $\text{tr } R$        | $9 = N_{\text{fam}}^2$ | $\text{tr } L$        | 15                                     |
| $\det R$              | $8 = h(D_5)$           | $\det L$              | $20 =  R(A_4)  = 2\mathcal{A}_\Lambda$ |
| $\chi_R(t)$           | $t^3 - 9t^2 + 10t - 8$ | $\chi_L(t)$           | $t^3 - 15t^2 + 40t - 20$               |
| $\text{PrinMin}_2(R)$ | $(2, 3, 5)$            | $\text{PrinMin}_2(L)$ | $(5, 14, 21)$                          |
| $\ R\ _F^2$           | $78 = \dim E_6$        | —                     | —                                      |

The three principal  $2 \times 2$  minors of  $R$  are  $(2, 3, 5)$  — sheet, family, carrier — with product  $2 \cdot 3 \cdot 5 = 30 = h(E_8)$  and sum  $10 = \mathcal{A}_\Lambda$ . The wrong down-branch  $\{1, 3, 4\}$  fails exactly these invariants.

**Bilinear anchor forms.** With  $\mathbf{1} = (1, 1, 1)^\top$  and  $a = (1, 1, 2)^\top$  (all verified):

| $R$               | $\mathbf{1}$  | $a$             | $L$               | $\mathbf{1}$                  | $a$                           |
|-------------------|---------------|-----------------|-------------------|-------------------------------|-------------------------------|
| $\mathbf{1}^\top$ | $22 = \sum R$ | $27 = 3^3$      | $\mathbf{1}^\top$ | $40 =  R(D_5) $               | $45 = \dim D_5$               |
| $a^\top$          | $32 = 2^5$    | $40 =  R(D_5) $ | $a^\top$          | $56 = \dim \mathbf{56}_{E_7}$ | $64 = \dim S^+ \cdot  \mu_4 $ |

The corner value  $a^\top L a = 64 = (16, 4)$  reads exactly one spinor-gluon sheet of the  $E_8$  branching. And the rank-one winding update shows up in the inverses:

$$R^{-1} \mathbf{1} = \frac{1}{4}(1, 1, -1)^\top, \quad L^{-1} \mathbf{1} = \frac{1}{10}(1, 1, -1)^\top$$

i.e. the residue response reads the glue denominator  $|\mu_4| = 4$ , the full-length response reads the decouple denominator  $\mathcal{A}_\Lambda = 10$ . [E]

### Anchor-plane Plücker coordinates: scattered hits become oriented areas [E]

The strongest anti-numerology tool is to read *orientation invariants* of the anchor plane  $\langle \mathbf{1}, a \rangle$  rather than loose integers. For  $M \in \{R, Q, K, L\}$  the left block  $B_M = (\mathbf{1}^\top M; a^\top M)$  (a  $2 \times 3$ ) and the right



block  $C_M = (M \mathbf{1} \ Ma)$  (a  $3 \times 2$ ) have the  $2 \times 2$  Plücker minors

|     | $\text{Pl}(\cdot)$ (left)                     | $\text{Pl}_R(\cdot)$ (right)                                         |
|-----|-----------------------------------------------|----------------------------------------------------------------------|
| $R$ | $(-6, 2, 14)$                                 | $(8, 12, 4) = (h(D_5),  R(A_3) ,  \mu_4 )$                           |
| $Q$ | $(3, 6, 3)$                                   | $(0, 4, 5) = (0,  \mu_4 , g_{\text{car}})$                           |
| $K$ | $(-1, 6, 4) = (-N_\Phi,  R^+(A_3) ,  \mu_4 )$ | $(12, 12, -2) = ( R(A_3) ,  R(A_3) , - \mathbb{Z}_2 )$               |
| $L$ | $(6, 26, 14)$                                 | $(20, 30, 10) = (2\mathcal{A}_\Lambda, h(E_8), \mathcal{A}_\Lambda)$ |

Three load-bearing flavor numbers are now single oriented areas, not coincidences:

$$\|\text{Pl}(K)\|_1 = 11, \quad \|\text{Pl}_R(K)\|_1 = \text{Pl}(L)_{13} = 26, \quad \sum \text{Pl}_R(L) = 60 = D_{\text{start}}$$

(11 is the mass-power anchor-plane norm; 26 the  $c_t/c_b$  denominator; 60 the  $E_8$  cascade start). The left-null-space response complements the right one above: the column sums of  $\det L \cdot L^{-1}$  and  $\det R \cdot R^{-1}$  are  $(-5, 1, 6)$  and  $(1, -5, 6)$ , the same magnitudes  $\{N_\Phi, g_{\text{car}}, |R^+(A_3)|\} = \{1, 5, 6\}$ . Finally the sector rows themselves are anchor-shifted:  $K_e - K_u = (1, 1, 2) = a$  and  $L_e - L_u = (1, 2, 3) = \text{Exp}(A_3)$  — leptons are up-quarks displaced by the family roots. All entries are exact [\[E\]](#); this is an audit layer, not a closure (the physical quark amplitudes stay [\[C\]](#)). [\[verification/v37\\_plucker\\_anchor.py\]](#)

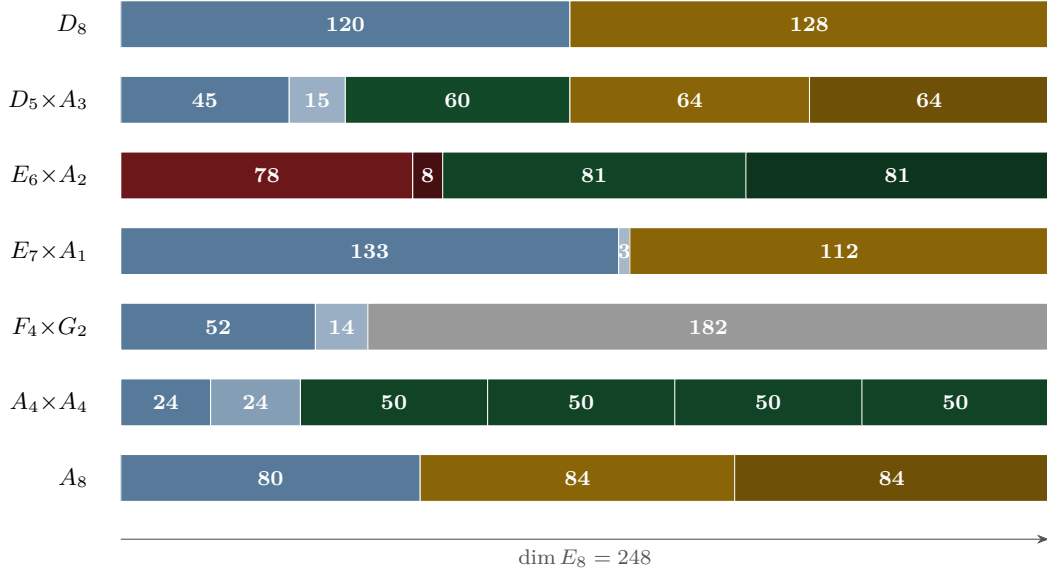


Figure 1: The  $E_8$  slice atlas: seven maximal-subalgebra cuts of 248. Gold = spinor-glue ( $64=(16, 4)$ ,  $112=7 \cdot 16$ ,  $84=7 \cdot 12$ ,  $128$ ); red = flavor-residue ( $78=\|R\|_F^2$ ,  $8=\det R$ ); green = family/action blocks; blue = Lie adjoints.

### The numerology null test: the look-elsewhere burden quantified [E]

The raster above disciplines *which* integers may appear; the complementary statistical question — *could a random “theory” of equal formula complexity land this many data hits?* — is answered by an explicit, fully declared null model. For each of the 13 scored frozen-registry observables the *entire* grammar class is enumerated exactly (no sampling): one-term expressions  $\frac{p}{q} (\varphi_0^{\text{ret}})^a \lambda_C^c (1+\lambda_C)^{d\pi^b} e^{u/w}$  with  $p, q \leq 130$ ,  $|a| \leq 2$ ,  $|c| \leq 3$ ,  $|d|, |b| \leq 1$ ,  $|u| \leq 6$ ,  $w \leq 6$ , and two-term sums with coefficients  $\leq 12$  — a class that provably contains every TFPT formula it scores (machine-checked fairness condition). Data windows are conservative: the *maximum* of the achieved deviation, the documented experimental  $\sigma$ , the quote resolution and the 5% quark scheme spread, so the known tension observables ( $\sin^2 \theta_{13}$ ,  $s_{23}$ ,  $\delta_{\text{CKM}}$ ,  $m_\mu/m_\tau$ ,  $\beta$ ) enter with their honest *large* windows and contribute almost nothing. Result of the census: joint formula-fishing probability  $\prod_i p_i = 10^{-25.8}$ , stable to  $< 2$  orders of magnitude when the coefficient bound is varied  $80 \dots 200$  and the plausibility window  $5 \times \dots 20 \times$ . Monte-Carlo cross-check (deterministic seeds):  $2 \times 2 \cdot 10^5$  pseudo-theories — random rational dials inside TFPT’s own formula skeletons, with the seed  $\varphi_0^{\text{ret}}$  fixed resp. drawn log-uniformly — reach at most 5/13 resp. 4/13 hits versus TFPT’s 13/13. The test has *power*: perturbing  $\varphi_0^{\text{ret}}$  by 1%/10%/30% collapses TFPT’s own scorecard to 9/6/4 hits, and shuffling which measured value is assigned to which formula collapses it to a mean of 1.2. Separately, all 94500 complexity-matched variants of the  $F_{U(1)}$  fixed-point equation (coefficients,  $M$ , transport exponent, resolvent power) are root-solved: exactly *one* — the TFPT instance  $(2, \frac{4}{5}, M=41, e^{-2\alpha}, -\frac{5}{4})$  — lands inside the achieved  $4 \cdot 10^{-8}$  CODATA window, so  $p_\alpha \leq 1.1 \cdot 10^{-5}$ . Headline:

$$P(\text{null reproduces the scorecard incl. } \alpha) \leq 10^{-30.7} \quad (\approx 102 \text{ bits})$$

*conditional on the declared grammar* — a look-elsewhere-corrected null-model rejection, never “certainty”. Complementary to the frozen registry: the freeze (v84) prevents *future* look-elsewhere drift, this test quantifies the *retrospective* look-elsewhere burden. [verification/v100\_numerology\_null\_mc.py]

## 4 The secondary branching atlas

Each maximal subalgebra of  $E_8$  mirrors a TFPT module. The dimension identities are exact [E]; the readings are audit-level [O]. The seven *slices* of  $\dim E_8 = 248$  are shown below: the same total cut seven ways, each cut a different TFPT module.

| Subalgebra       | 248 =                  | TFPT reading                                                                                                          |
|------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------|
| $D_8$            | 120 + 128              | family-transition adjoint + two spinor-gluon sheets<br>( $128=2 \dim S^+ \mu_4 $ )                                    |
| $D_5 \times A_3$ | 45 + 15 + 60 + 64 + 64 | <b>main split:</b> $\dim D_5, \dim A_3, \mathcal{A}_\Lambda  R^+(A_3) $ , two (16, 4) glue sheets                     |
| $E_6 \times A_2$ | 78 + 8 + 81 + 81       | <b>flavor residue:</b> $\ R\ _F^2, \det R = \dim A_2$ , cubic family blocks                                           |
| $E_7 \times A_1$ | 133 + 3 + 112          | <b>scalaron:</b> $112=7 \cdot 16$ (exponent $\times$ one family)                                                      |
| $F_4 \times G_2$ | 52 + 14 + 182          | <b>occupancy/gauge:</b> $ R(F_4) =48=\Omega_{\text{adm}},  R(G_2) =12=\dim \mathfrak{g}_{\text{SM}}$                  |
| $A_4 \times A_4$ | 24 + 24 + 4 · 50       | <b>action ladder:</b> $24+24=48=\Omega_{\text{adm}}$ ; off-diagonal (5, 10) reps $\mathcal{A}_H, \mathcal{A}_\Lambda$ |
| $A_8$            | 80 + 84 + 84           | rank-decuplet + scalaron: $80=8 \cdot 10, 84=7 \cdot 12$                                                              |

| The $D_8$ split is a magnitude/phase channel [E]/[C]/[O]<br>( [verification/v227_degree_exponent_channel_split.py] )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The $D_8$ row $248 = 120 + 128$ is the canonical <i>exponent/degree</i> split of $E_8$ : the exponents sum to $120 =  R^+(E_8) $ (a <b>magnitude</b> channel — positive roots, root transitions, the dimensionless ratio/product readouts), the fundamental degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ sum to $128 = 2^{\text{rank}-1} = \text{rank } E_8 \cdot \dim S^+$ (a <b>phase/gluon</b> channel — the spinor/Ramond sheets). This gives the red-team Target D CP residual a clean home: the (ratios, product) $\Rightarrow v_{\text{geo}}$ magnitude bijection lives in the 120 channel, the un-covered CP phases in the 128 channel (cf. the hexagonal phase fiber, red team Target D). Whether 128 also hosts a dark sector is Frontier [O] — this is the honest replacement for the over-strong “ $S^-$ is dark matter” reading, not an assertion of it. |

| $E_8$ Slice Compression Lemma [E] (audit)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The seven slices are not seven separate narratives — they are <i>one</i> projection of two small alphabets. The anchor power sums $p_n(a) = 2 + 2^n$ give $P = (p_0, p_1, p_2, p_3) = (3, 4, 6, 10)$ , whose six $K_4$ edge products are exactly the carrier blocks $(p_0 p_1, \dots, p_2 p_3) = (12, 18, 30, 24, 40, 60) = ( R(A_3) , N_{\text{fam}}  R^+(A_3) , h(E_8),  W(A_3) ,  R(D_5) , D_{\text{start}})$ ; the six Sheet-Diamond operators carry determinants $(\det Q, \det K, \det R, \det C, \det L, \det F) = (3, 4, 8, 14, 20, 32)$ with $\sum \det = 81 = N_{\text{fam}}^4$ , the total determinant charge of the flavour space. All seven 248-slices, the row-budget cross $\text{row}(C + sU + tV) = (7, 11, 13) + s(3, 3, 3) + t(1, 2, 3)$ (democratic winding + $A_3$ exponents), and $\dim E_6 = 78 = p_2(p_0 + p_3)$ then follow from $P$ , those determinants and $\Delta = p_0 + p_3 = 13$ . The atlas is therefore a closed grammar over admissible invariant classes (power sums, determinants, branching dimensions), not a shelf of trophies ( [verification/v170_diamond_slice_compression.py] ). |

**Theorem 1** ( $E_6 \times A_2$  flavor-residue shadow [O]). The principal flavor branching reads the residue matrix:

$$\|R\|_F^2 = 78 = \dim E_6, \quad \det R = 8 = \dim A_2 = h(D_5), \quad \mathbf{1}^\top R a = 27,$$

and the full  $E_6 \times A_2$  dimension count is the residue identity

$$248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\text{fam}} = 78 + 8 + 2 \cdot 27 \cdot 3.$$

Here  $E_6$  reads the Frobenius norm of the residue matrix,  $A_2$  the pure three-family symmetry, and the **27** the cubic family block. (Strongest new  $E_8$  flavor fingerprint.)

**Theorem 2** ( $E_7 \times A_1$  scalaron shadow [O]). With the scalaron exponent  $7 = \Omega_{\text{adm}} - 10b_1 = -\deg E + \text{rk } E$ ,

$$248 = 133 + 3 + 112, \quad 112 = 7 \cdot \dim S^+ = 7 \cdot 16, \quad |R(E_7)| = 126 = 7 \cdot 18,$$

where  $18 = N_{\text{fam}} |R^+(A_3)|$ . The scalaron exponent 7 (the seam power  $M_{\text{scal}}^2 / \bar{M}_{\text{P1}}^2 = c_3^7$ ) is thus exactly the  $E_7$  off-block coefficient. *Cross-check:* the centred  $E_8$  exponents give  $\sum_{m \in \text{Exp}(E_8)} |m - 15| = 56 = \dim \mathbf{56}_{E_7}$  (exponents  $\{1, 7, 11, 13, 17, 19, 23, 29\}$ , the primitive residues mod 30).

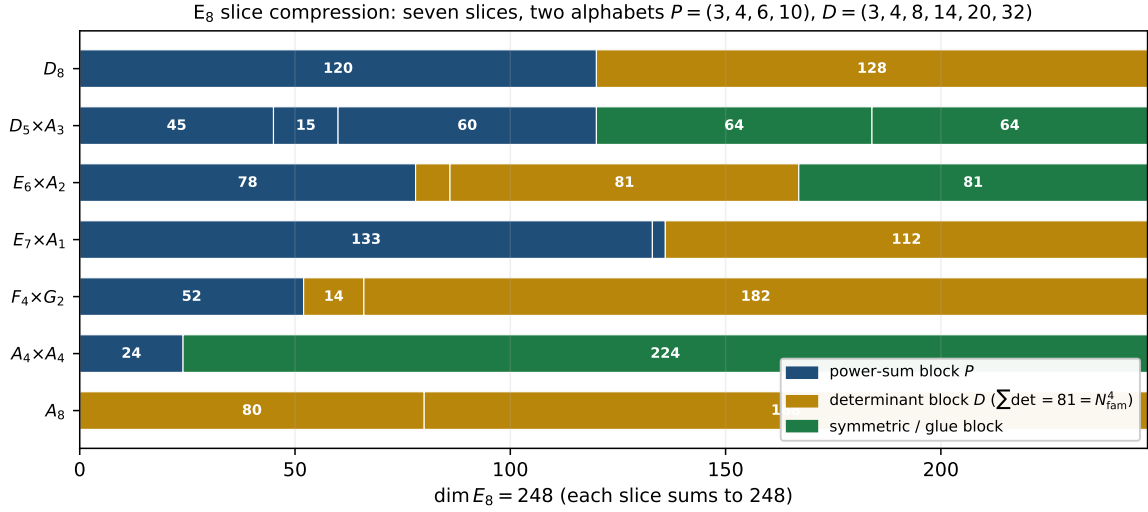


Figure 2: The same seven 248-slices, now coloured by their *alphabet* source rather than by physics module: blue = anchor power-sum blocks ( $P = (3, 4, 6, 10)$ ), gold = Sheet-Diamond determinant blocks ( $D = (3, 4, 8, 14, 20, 32)$ ,  $\sum \det = 81 = N_{\text{fam}}^4$ ), green = symmetric/glue data. Every slice is one projection of the two alphabets ([[verification/v170\\_diamond\\_slice\\_compression.py](#)]).

**Proposition 1** ( $A_4 \times A_4$  action-ladder and  $F_4 \times G_2$  occupancy [O]).  $248 = 24 + 24 + 4(5 \cdot 10)$  with  $24 + 24 = 48 = \Omega_{\text{adm}}$  and off-diagonal reps  $(\mathbf{5}, \mathbf{10}) = (\mathcal{A}_H, \mathcal{A}_\Lambda)$  — the action ladder appears inside  $E_8$  (an audit, *not* an  $SU(5)$  GUT). And  $F_4 \times G_2$ :  $|R(F_4)| = 48 = \Omega_{\text{adm}}$ ,  $|R(G_2)| = 12 = \dim \mathfrak{g}_{\text{SM}}$ ,  $\text{rank } F_4 = 4 = |\mu_4|$ ,  $\text{rank } G_2 = 2 = |\mathbb{Z}_2|$ .

**Flavor-matrix and prime-table fingerprints (audit-level, machine-verified).** Five exact linear-algebra readouts of  $R, L$  and the prime atoms, kept here under the no-free-pattern rule.

- **Row norms ladder.**  $\|R_u\|^2, \|R_d\|^2, \|R_e\|^2 = (10, 30, 38) = (\mathcal{A}_\Lambda, h(E_8), 2 \cdot 19)$  with  $10+30+38 = 78 = \dim E_6$  ( $19 \in \text{Exp}(E_8)$ ).
- **Residue inventory.** The entries of  $R$  on  $C_6$  are  $m = (1, 2, 2, 2, 0, 2)$ : the *only* gap is  $4 = |\mu_4| = \sum a$ . Moments  $\sum R = 22$ ,  $\sum R^2 = 78 = \dim E_6$  — so  $\|R\|_F^2 = 78$  is the second moment of the  $C_6$  inventory with the  $\mu_4$  hole.
- **First-column norm**  $\rightarrow E_6 \times A_2$  **off-blocks.**  $L_{\cdot 1} = (7, 7, 8)$ ,  $\|L_{\cdot 1}\|^2 = 162 = 81 + 81$  — the two **81** off-blocks of the  $E_6 \times A_2$  slice are exactly the wound first-generation column.
- **Lepton exponent polynomial.** The  $\varphi_0^{\text{ret}}$ -ladder lepton powers  $(k_e, k_\mu, k_\tau) = (5, 3, 2)$  are the Coxeter primes  $30 = 2 \cdot 3 \cdot 5$  in hierarchy order:  $(t-5)(t-3)(t-2) = t^3 - 10t^2 + 31t - 30$  ( $e_1 = \mathcal{A}_\Lambda$ ,  $e_3 = h(E_8)$ ). The lepton  $c$ -ratios obey  $\frac{(16/7)(7/6)}{4/3} = 2 = |\mathbb{Z}_2|$  and  $\frac{(16/7)(7/6)}{(4/3)^2} = \frac{3}{2}$  (the near-Koide factor).
- **The 2, 3, 5 multiplication table** (the shortest integer-landscape compression):

$$(2, 3, 5) \xrightarrow{\times} (6, 10, 15, 30) \rightarrow (8=3+5, 16, 240=8 \cdot 30, 248=240+8).$$

## 5 Group-theoretic audits (optional)

**Weyl completion index.** Over the main stabiliser,

$$\frac{|W(E_8)|}{|W(D_5)| |W(A_3)|} = \frac{696729600}{1920 \cdot 24} = 15120 = 63 \cdot 240 = (2^6 - 1) |R(E_8)|.$$

A group-theory machine-check, not physics: it quantifies the Weyl symmetry created beyond the  $D_5 \times A_3$  stabiliser. [E] (arithmetic) / [O] (reading)

$E_8$  **theta series as a higher-shell raster.**  $\Theta_{E_8} = E_4 = 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n$  has shell degeneracies 240, 2160, 6720, 17520, ... The first is  $|R(E_8)| = 240$ ; the second is  $2160 = 9 \cdot 240 = N_{\text{fam}}^2 |R(E_8)|$ . If later one-loop or CMB shell degeneracies touch these multiplicities, the theta series is the right place to type them — as an audit raster, not a claim. [O]

## 6 Further audit lemmas (machine-verified this round)

Five more exact identities tie the flavor matrices and the  $E_8$  exponents to the compiler. All checked with exact integer arithmetic.

(i) **Full-length Frobenius norm is three  $E_6$  blocks.**  $\|L\|_F^2 = 234 = N_{\text{fam}} \|R\|_F^2 = 3 \dim E_6$ , and  $\|L\|_F^2 = \|R\|_F^2 + 12\langle R, W \rangle + 36\|W\|_F^2 = 78 + 48 + 108$ , with  $\langle R, W \rangle = 4 = |\mu_4|$  and  $\|W\|_F^2 = 3 = N_{\text{fam}}$ . The full word-length load is exactly the residue  $E_6$  norm + the  $A_3$ -winding over  $\mu_4$  + the family-winding square. [E]

(ii)  **$A_3$ -winding determinant lemma.** The rank-one winding  $L = R + 6\mathbf{1}e_1^\top$  and the matrix-determinant lemma with  $e_1^\top R^{-1}\mathbf{1} = \frac{1}{4}$  give  $\det L = \det R(1 + \frac{|R^+(A_3)|}{|\mu_4|}) = 8(1 + \frac{6}{4}) = 20 = 2\mathcal{A}_\Lambda$ .

So  $h(D_5) = 8 \xrightarrow{A_3^+/\mu_4} 20 = |R(A_4)|$ : the winding injects the carrier factor 5 into the cokernel ( $\text{coker } R \simeq \mathbb{Z}_8$ ,  $\text{coker } L \simeq \mathbb{Z}_{20}$ ). [E]

(iii) **The scalaron 7 is the  $E_8$  exponent variance per decuple.** Centring the exponents at  $h/2 = 15$ ,  $\sum_m (m - 15)^2 = 560$  and  $\text{Var} = \frac{560}{8} = 70 = 7 \cdot \mathcal{A}_\Lambda = (\text{scalaron } 7) \cdot (\text{decuple } 10)$ . The halved positive offsets  $\{1, 2, 4, 7\}$  satisfy  $\sum = 14 = 2 \cdot 7$ ,  $\sum^2 = 70$ ,  $\prod = 56 = \dim \mathbf{56}_{E_7}$  — a micro-code carrying the scalaron, the decuple and the  $E_7$  minuscule at once. Also  $\sum_{\text{pairs}} m(30 - m) = 620 = \frac{h \dim E_8}{12} = 2 \cdot 10 \cdot 31$ . [E]

(iv) **Exponents + degrees give the 120+128 split.** The invariant degrees  $d_i = m_i + 1 = \{2, 8, 12, 14, 18, 20, 24, 30\}$  obey  $\sum_i m_i = 120$ ,  $\sum_i d_i = 128$ ,  $248 = 120 + 128$ ,  $\prod_i d_i = |W(E_8)|$ , matching the  $D_8$  branching  $248 = 120 + 128$ ; the degrees are a TFPT dictionary (2, 8, 12, 14, 18, 20, 24, 30 = sheet,  $h(D_5)$ ,  $|R(A_3)|$ ,  $\dim G_2$ ,  $p_4$ ,  $\det L$ ,  $\dim A_4$ ,  $h(E_8)$ ). [E]

(v) **Root shell = local  $E_7 \times A_1$ .** Around any  $E_8$  root the inner-product multiplicities are  $\{+2:1, +1:56, 0:126, -1:56, -2:1\}$ , i.e. the local  $248 = (133, 1) + (1, 3) + (56, 2)$ . A distinguished root (seam/Higgs direction) leaves an orthogonal  $E_7$  and an active coupling shell  $56+56$ . [E]

## 7 What this changes, and what stays open

$E_8$  is one container whose maximal subalgebras each project a different TFPT module:

**Net.** The atlas turns isolated fingerprints (e.g.  $\|R\|_F^2 = 78 = \dim E_6$ , the scalaron 7) into *branching projections* of one container. The discipline rule — “every load-bearing number must appear in  $\geq 1$  projection” — makes the structure falsifiable: a number in no projection is suspect. This is a *program*, not a proof of new physics: all readings are [O]. The two genuinely load-bearing, derived results remain the  $\mu_4$  glue ( $E_8 = (D_5 \oplus A_3) + \mu_4$ , [E] [verification/v1\_e8\_glue.py]) and the spectral selector of the flavor matrix ( $\det R = h(D_5)$ ,  $\text{PrinMin}_2 = (2, 3, 5)$ , [E] [verification/v4\_flavor\_matrix.py]). The remaining flavor-geometry step (historically “H2”) is sharpened in the parabolic note to: *the  $D_4$ -equivariant stable parabolic structure on  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$  realises the unique branch with  $\det R = 8$  and  $\text{PrinMin}_2(R) = (2, 3, 5)$  — no longer “find the matrix”, but “prove this one geometric realisation”.* (Status: since v118–v122 the dictionary *values* are exact theorems — hexagon = sign-twisted family spectrum, addresses pinned, margins forced; what remains is exactly the realisation statement above, the selector establishment  $R4'$ .)

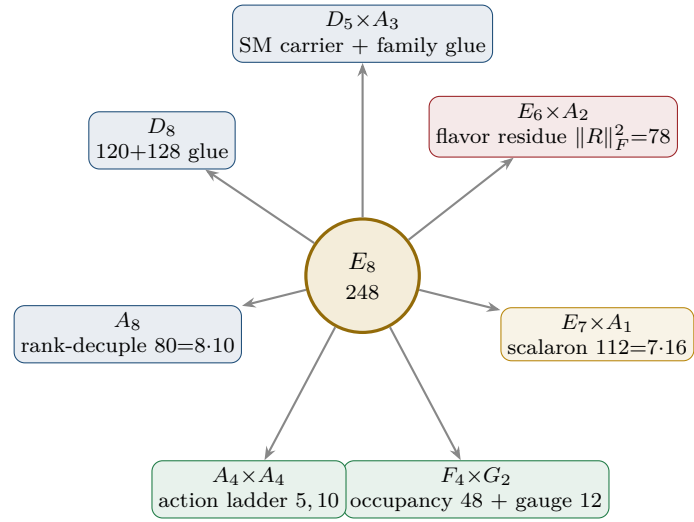


Figure 3:  $E_8$  as an audit container: each maximal subalgebra projects one TFPT module. The discipline rule — every load-bearing number appears in  $\geq 1$  projection — makes the structure falsifiable.

## Part II

# The $E_8$ cascade bridge

**The connection in one sentence.** The old  $E_8$  cascade (Paper V1.06: the nilpotent-orbit chain  $D_n = 60 - 2n$ ,  $n = 0 \dots 26$ ) and the new compiler/atlas are *the same even-integer  $E_8$  spine*, read three ways: the cascade *orders* the numbers  $\{8, 10, \dots, 60\}$  as a scale ladder, the atlas *reads* them as  $E_8$  subalgebra branchings, and the compiler *generates* them from the  $(1, 1, 2)$  anchor. The cascade *starts* at  $D_{\text{start}} = 60 = p_2 p_3$  and *ends* at  $8 = h(D_5) = \det R$  — the flavor selector.

## 8 The old cascade, recalled

Paper V1.06 builds a deterministic scale ladder from the nilpotent orbits of  $E_8$ . With centralizer dimension  $D = 248 - \dim \mathcal{O}$  and the Hasse chain restricted to  $\Delta D = 2$ , a unique maximal chain runs from  $A_4 + A_1$  ( $D_0 = 60$ ) down to the regular orbit ( $D_{26} = 8$ ):

$$D_n = 60 - 2n, \quad n = 0, \dots, 26; \quad \varphi_{n+1} = \varphi_n e^{-\gamma(n)}, \quad \gamma(n) = \lambda [\ln D_n - \ln D_{n+1}]$$

The cascade arithmetic (endpoints, exponent rungs, IR-tail product, variance) is machine-checked. [\[verification/v5\\_e8\\_cascade.py\]](#) with the single structural normalisation  $\lambda = \gamma(0)/(\ln 248 - \ln 60) = 0.58770$ , fixed by a curvature-extremum on the chain (no fit). The  $E$ -windows of the two-loop flow anchor it:  $\alpha_3 = 1/(8\pi) = c_3$  (the  $E_8$  window) and  $\alpha_3 = 1/(6\pi)$  (the  $E_6$  window, near  $\varphi_0^{\text{ret}}$ ).

## 9 The spine: every rung is a current number

Reading  $D = 60 - 2n$  against today's compiler atoms, the secondary-branching atlas and the  $(1, 1, 2)$  anchor power sums  $p_n = 2 + 2^n$ , **25 of the 27 rungs are load-bearing numbers** (machine-checked):

| $n$ | $D$ | current meaning                                                                | $n$ | $D$ | current meaning                                          |
|-----|-----|--------------------------------------------------------------------------------|-----|-----|----------------------------------------------------------|
| 0   | 60  | $D_{\text{start}} = p_2 p_3 = \mathcal{A}_\Lambda  R^+(A_3) $ ( <b>start</b> ) | 13  | 34  | $p_5$ anchor power = $2 \cdot 17$                        |
| 1   | 58  | $2 \cdot 29$ ( $E_8$ exponent)                                                 | 14  | 32  | $2^5 = 2 \dim S^+$                                       |
| 2   | 56  | $\dim \mathbf{56}_{E_7} = a^\top L \mathbf{1} = \sum  m-15 $                   | 15  | 30  | $h(E_8) = 2 \cdot 3 \cdot 5$ ( <b>Coxeter rung</b> )     |
| 3   | 54  | $2 \cdot 27$ , $27 = \mathbf{1}^\top R a = 3^3$                                | 16  | 28  | $4 \cdot 7$ (half of 56)                                 |
| 4   | 52  | $\dim F_4 =  R(D_5)  +  R(A_3) $                                               | 17  | 26  | $2 \cdot 13$ (old hadronic block)                        |
| 5   | 50  | $A_4 \times A_4$ block = $5 \cdot 10$                                          | 18  | 24  | $\dim A_4$ ; $24+24=48$                                  |
| 6   | 48  | $\Omega_{\text{adm}} =  R(F_4)  = 2p_1 p_2$                                    | 19  | 22  | $\sum R = \mathbf{1}^\top R \mathbf{1}$ (residue sum)    |
| 7   | 46  | $2 \cdot 23$ ( $E_8$ exponent)                                                 | 20  | 20  | $\det L =  R(A_4)  = 2\mathcal{A}_\Lambda$               |
| 8   | 44  | $4 \cdot 11$ ( $E_8$ exponent)                                                 | 21  | 18  | $p_4 = N_{\text{fam}}  R^+(A_3) $                        |
| 9   | 42  | $6 \cdot 7 =  R^+(A_3)  \cdot 7$                                               | 22  | 16  | $\dim S^+$ (one generation)                              |
| 10  | 40  | $ R(D_5)  = \sum L = \mathbf{1}^\top L \mathbf{1}$                             | 23  | 14  | $\dim G_2 = \sum L_d = 8+6$                              |
| 11  | 38  | $2 \cdot 19$ ( $E_8$ exponent)                                                 | 24  | 12  | $ R(A_3)  = \dim \mathfrak{g}_{\text{SM}}$               |
| 12  | 36  | $6^2$ (old EW block)                                                           | 25  | 10  | $\mathcal{A}_\Lambda = \binom{5}{2}$ (old cosmo block)   |
|     |     |                                                                                | 26  | 8   | $h(D_5) = \det R = 2p_1$ ( <b>end: flavor selector</b> ) |

Three structural facts jump out (all [\[E\]](#) arithmetic):

- **Endpoints are the compiler boundary.** The cascade *starts* at  $60 = p_2 p_3 = D_{\text{start}}$  and *ends* at  $8 = h(D_5) = \det R$  — the flavor spectral selector is literally the deepest orbit of  $E_8$ .
- **Coxeter rung is  $h(E_8)$ .** At  $n = 15$ ,  $D = 30 = h(E_8)$  — this is the *Coxeter rung*, not the arithmetic centre; the median node is  $n = 13$ ,  $D = 34 = p_5(a)$  (below). It matches the exponent-centering audit  $\sum_m |m - 15| = 56 = \dim \mathbf{56}_{E_7}$  (which is itself rung  $n = 2$ ).
- **Doubled  $E_8$  exponents are rungs.**  $\{58, 46, 38, 34, 26, 22, 14\} = 2 \times \{29, 23, 19, 17, 13, 11, 7\}$  are exactly the cascade rungs at  $n = 1, 7, 11, 13, 17, 19, 23$ ; three of them coincide with anchor

numbers ( $34 = p_5$ ,  $22 = \sum R$ ,  $14 = \dim G_2$ ). Only  $n = 8$  ( $D = 44 = 4 \cdot 11$ ) is not independently labelled.

### Cascade arithmetic — endpoints, centre and length (verified)

The 27 rungs encode  $E_8$  in their endpoints, and their centre is the anchor:

$$\begin{aligned} \frac{D_{\text{start}} D_{\text{end}}}{2} &= \frac{60 \cdot 8}{2} = 240 = |R(E_8)|, & 240 + D_{\text{end}} &= 248 = \dim E_8, & D_{\text{start}} &= 2h(E_8) = 60; \\ \text{arithmetic median} &= \frac{60+8}{2} = 34 = p_5(a) = h(E_8) + |\mu_4|, & \sum_{n=0}^{26} (D_n - 30) &= 27 \cdot 4 = 27|\mu_4|; \\ \text{width} &= D_{\text{start}} - D_{\text{end}} = 52 = \dim F_4, & 60 \rightarrow 30 &= h(E_8), & 30 \rightarrow 8 &= \sum R = 22; \\ \#\text{nodes} &= 27 = 3^3 = \mathbf{1}^\top Ra \text{ (} E_6 \text{ cubic block)}, & \#\text{steps} &= 26 = \dim \mathbf{26}_{F_4}. \end{aligned}$$

So the cascade is an  $E_8$  Coxeter ladder centred on  $p_5(a)$  with a per-node  $|\mu_4|$  glue shift; its length (27 nodes, 26 steps) carries the  $E_6$  cube and the  $F_4$  fundamental. [E]

### The IR tail $n = 24 \dots 30$ : the compiler atoms

Continuing  $D = 60 - 2n$  past the last orbit ( $D=8$ ,  $n=26$ ) is no longer a nilpotent-orbit chain but the *algebraic compiler tail* of the Coxeter clock:

$$D_{24 \dots 30} = \underbrace{12}_{|R(A_3)|=\dim \mathfrak{g}_{\text{SM}}}, \underbrace{10}_{\mathcal{A}_\Lambda}, \underbrace{8}_{h(D_5)}, \underbrace{6}_{|R^+(A_3)|}, \underbrace{4}_{|\mu_4|}, \underbrace{2}_{\text{sheet}}, \underbrace{0}_{\text{closure}}.$$

This is exactly what the old “ $n = 0 \dots 30$  clock” was scanning:  $n=0 \dots 26$  the  $E_8$  orbit spine,  $n=27 \dots 30$  the compiler atoms. (Mark the tail as the *compiler tail*, not orbits.) [E]



**Three further cascade lemmas (all machine-verified)**

The clock carries more  $E_8$  data than the endpoints:

- **Primitive rungs sum to the root count.** Evaluating  $D_n = 60 - 2n$  at the *primitive* indices  $n \in (\mathbb{Z}/30)^\times = \{1, 7, 11, 13, 17, 19, 23, 29\} = \text{Exp}(E_8)$  gives  $\{58, 46, 38, 34, 26, 22, 14, 2\}$  with

$$\sum_{n \in \text{Exp}(E_8)} (60 - 2n) = 240 = |R(E_8)|,$$

and since  $30 - n$  is primitive when  $n$  is, the clock indexes its own  $E_8$  exponents.

- **The IR tail product is the main Weyl stabiliser.** The nonzero tail  $\{12, 10, 8, 6, 4, 2\}$  has

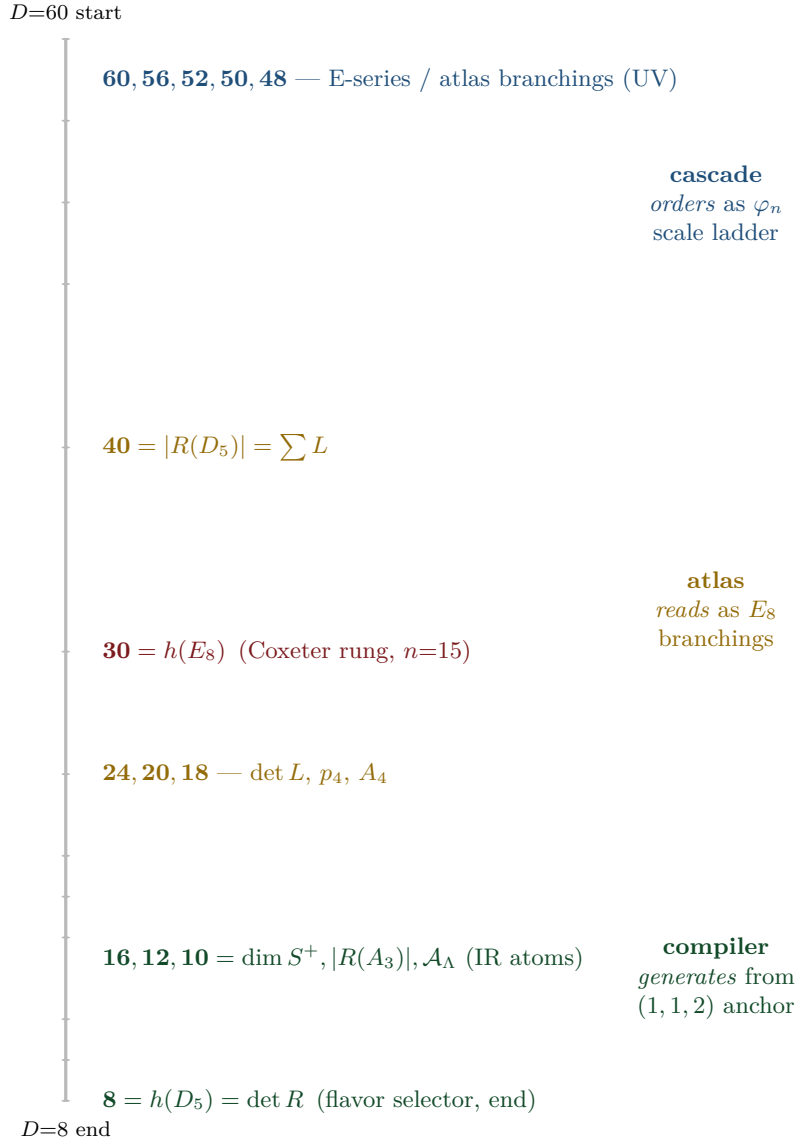
$$\prod = 46080 = |W(D_5)| |W(A_3)| = 1920 \cdot 24, \quad \sum = 42 = |R^+(A_3)| \cdot 7.$$

- **Cascade variance =  $E_6$  flavour norm times the scalaron-gauge block.** The 27 rungs  $60, \dots, 8$  have mean  $34 = p_5(a)$  and

$$\sum_{n=0}^{26} (D_n - 34)^2 = 6552 = 78 \cdot 84 = \dim E_6 \cdot (7 \dim \mathfrak{g}_{\text{SM}}),$$

tying the cascade simultaneously to the  $E_6 \times A_2$  flavour shadow and the  $A_8$  scalaron-gauge shadow.  
**[E]** (audit level)

## 10 Three views of one spine



**The unification.** The same numbers  $\{8, 10, \dots, 60\}$  are: (a) the *scale ladder*  $\varphi_n$  of the old cascade (orbit dimensions, ordered by  $e^{-\gamma}$ ); (b) the *branching blocks* of the secondary-atlas ( $\dim F_4=52$ ,  $\dim \mathbf{56}_{E_7}, 5 \cdot 10, \dots$ ); (c) the *anchor power sums and matrix invariants* of the compiler ( $p_n = 2 + 2^n$ ,  $\det R=8$ ,  $\sum L=40$ ). The cascade is the *UV*→*IR* ordering; the compiler is its *IR endpoint*, where the small numbers 8, 10, 12, 16 are the flavor/family/carrier data.

**The spine tetrahedron: the microkernel as one finite object [E]**  
( [verification/v91\_spine\_tetrahedron.py] )

The spine  $T = \{2, 3, 4, 5\} = \{\mathbb{Z}_2, N_{\text{fam}}, |\mu_4|, g_{\text{car}}\}$  equals  $\{e_3(a), p_0(a), e_1(a), e_2(a)\}$  for the anchor  $a = (1, 1, 2)$  — the tetrahedron on these four nodes is the anchor plus its zeroth moment, not a second structure. Its combinatorics *are* the central integer grammar: edge products  $\{6, 8, 10, 12, 15, 20\}$  ( $|R^+(A_3)|$ ;  $\text{rank } E_8=h(D_5)=\det R$ ;  $\mathcal{A}_\Lambda=|E(K_5)|$ ;  $|R(A_3)|=\dim \mathfrak{g}_{\text{SM}}$ ;  $\dim A_3$ ;  $\det L$ ), face products  $\{24, 30, 40, 60\}$  ( $|W(A_3)|$ ;  $h(E_8)$ ;  $|R(D_5)|=\sum L$ ;  $D_{\text{start}}$ ), volume  $2 \cdot 3 \cdot 4 \cdot 5 = 120 = |R^+(E_8)| = 5!$ ; and the graph reading  $|R(E_8)| = 240 = |\mu_4| \cdot |E(K_4)| \cdot |E(K_5)|$  with  $|E(K_4)| = p_2 = 6$ ,  $|E(K_5)| = p_3 = 10$  — including the honest negative control that the identification *breaks* at  $K_6$  ( $|E(K_6)| = 15 \neq p_4 = 18$ ), so it is specific, not generic. Two disciplined caveats: the “dual cuts” (node  $\times$  opposite face = 120, etc.) are tautologies of any complementary 4-set partition and are typed as presentation only; and the tetrahedron grammar is a *sub*-grammar — the load-bearing 7, 16, 41, 48, 240, 248 are *not* subset products and still require the

anchor power sums / Pascal / glue layers.

## 11 Direct $\varphi_0^{\text{ret}}$ relations: old vs current

The old cascade’s “fundamental relations near  $n = 0$ ” connect directly to the current note — one is *identical*, two are early forms of current refinements:

| Quantity        | old (V1.06)                                  | current                                                                          | relation                                                |
|-----------------|----------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------|
| $\Omega_b$      | $\varphi_0^{\text{ret}}(1 - 2c_3) = 0.04894$ | $(1 - \frac{1}{4\pi})\varphi_0^{\text{ret}} = 0.04894$                           | <b>identical</b> ( $2c_3 = \frac{1}{4\pi}$ ) [E]        |
| $V_{us}/V_{ud}$ | $\sqrt{\varphi_0^{\text{ret}}} = 0.2306$     | $\lambda_C = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})} = 0.2244$ | refined (the $(1 - \varphi_0^{\text{ret}})$ factor) [E] |
| $r$ (tensor)    | $(\varphi_0^{\text{ret}})^2 = 0.00283$       | $12/N_\star^2 = 0.00397$                                                         | two branches, same order [C]                            |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

So  $\Omega_b = \varphi_0^{\text{ret}}(1 - 2c_3)$  was already the exact current formula in V1.06; the Cabibbo ratio gained its  $(1 - \varphi_0^{\text{ret}})$  factor; and the old direct  $r = (\varphi_0^{\text{ret}})^2$  and the current Starobinsky  $r = 12/N_\star^2$  agree in order ( $\sim 0.003$ ) but are distinct predictions — a clean, testable discriminant, stated honestly.

## 12 Sector inventory: have all the old-cascade sectors been computed?

The old cascade also carried *calibrated absolute scales*: per block, one unit  $\zeta_B$  was fixed on a single reference and every other scale in the block followed fit-free from the ratio law  $\varphi_m/\varphi_n = (D_m/D_n)^\lambda$  (Paper V1.06, App. C). These block assignments are the place where the proton mass, the EW masses and the CMB temperature lived. The honest question is: *has the new compiler re-derived each of these, or only some?* The full inventory, with the current status spelled out:

| Old block (rung)     | Old observable & calibration                                                                                                             | Current-model status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Mark |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| near $n=0$           | $\Omega_b = \varphi_0^{\text{ret}}(1 - 2c_3),$<br>$V_{us}/V_{ud} = \sqrt{\varphi_0^{\text{ret}}},$<br>$r = (\varphi_0^{\text{ret}})^2$   | <b>covered</b> — dimensionless, derived from $\{c_3, \varphi_0^{\text{ret}}\}$ ; see the $\varphi_0^{\text{ret}}$ -table above. $\Omega_b$ is <i>identical</i> .                                                                                                                                                                                                                                                                                                                                                        | [E]  |
| EW, $n=12$           | $v_H = \zeta_{\text{EW}} M_{\text{P}1} \varphi_{12},$<br>$M_W = \frac{1}{2} g_2 v_H, M_Z$                                                | <b>scale covered, absolute scheme-level</b> — the EW scale enters the current note via the action-grammar exponent and the two-loop flow; $M_W, M_Z$ are dimensionful and remain RG/scheme outputs (one $\zeta_{\text{EW}}$ ), <i>not</i> new compiler powers.                                                                                                                                                                                                                                                          | [C]  |
| hadronic, $n=15, 17$ | $m_p = \zeta_p M_{\text{P}1} \varphi_{15},$<br>$m_b = \zeta_b M_{\text{P}1} \varphi_{15},$<br>$m_u = \zeta_u M_{\text{P}1} \varphi_{17}$ | <b>ratios closed, absolute scale an anchor.</b> The quark mass <i>ratios</i> sit on the $\varphi_0^{\text{ret}}$ -word-length ladder and are closed (integer Plücker readouts, <code>v49/v71</code> ); the <i>absolute</i> values need the per-sector $\Lambda/U_f^\star$ normalisation, an anchor ([O]). The <b>proton mass</b> is QCD-confinement, $m_p \sim \Lambda_{\text{QCD}}$ by dimensional transmutation — a <i>cross-sector</i> ratio, listed as a genuine frontier item ( $m_p/m_e$ , <code>tfpt_4</code> ). | [O]  |
| cosmo, $n=25$        | $T_{\gamma 0} = \zeta_\gamma M_{\text{P}1} \varphi_{25},$<br>$T_\nu = (\frac{4}{11})^{1/3} T_{\gamma 0}, \rho_\Lambda$                   | <b>partly.</b> $\rho_\Lambda/\Lambda$ is <i>typed</i> by the action grammar and the seam ( $\Lambda$ -typing, <code>tfpt_2</code> ); $T_\nu/T_{\gamma 0}$ is standard QFT (not cascade). The <b>absolute CMB temperature</b> $T_{\gamma 0}$ is a downstream readout of the reheating pipeline (the old cosmology/CMB layers; today the scalaron-reheating chain, <code>v86</code> ), conditional — not re-derived as a compiler number.                                                                                 | [C]  |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

**Direct answer: which sectors are closed, which are not**

**Closed (derived from  $\{c_3, g_{\text{car}}, \varphi_0^{\text{ret}}\}$ ):** the dimensionless near- $n=0$  anchors ( $\Omega_b$  exactly, Cabibbo, tensor branch) and all the *dimension/rung* structure of the spine. **Scale-level only (one  $\zeta_B$  each, scheme/RG):** the EW masses  $M_W, M_Z$  and the cosmological absolute scales  $T_{\gamma 0}$  — exactly as in the current SM/cosmology notes; these were *never* pure predictions, not even in V1.06 (each needed a block unit). **Genuinely open:** the **proton mass**  $m_p$  (and  $m_p/m_e$ ) is QCD-confinement across two sectors and is *not* claimed as a compiler power — it stays a frontier item ([O], tfpt\_4); the quark mass *ratios* are closed (integer Plücker,  $\sqrt{49}/\sqrt{71}$ ) and only the *absolute* quark scale  $m_b, m_u$  needs the per-sector  $\Lambda/U_f^*$  normalisation, an anchor ([O]). So: yes, the *spine* is fully audited and the *dimensionless* sectors are derived; the *dimensionful* hadronic/cosmo absolute scales are honestly still scheme-level, with the proton mass the one explicitly open cross-sector number — nothing is silently claimed.

## 13 What is new, what is old, what stays open

**New structures found via the bridge**

1. The flavor selector  $\det R = h(D_5) = 8$  is the *terminal orbit* of the  $E_8$  cascade; the compiler is the cascade's IR endpoint. [E]
2. The atlas branching blocks (52, 56, 60) are the cascade's *UV head* ( $n = 0, 2, 4$ ); the secondary atlas and the orbit cascade are the same spine read top-down vs. as subalgebras. [E]
3. The anchor power sums sit on doubled-exponent rungs:  $p_5 = 34$  ( $n=13$ ),  $\sum R = 22$  ( $n=19$ ),  $\dim G_2 = 14$  ( $n=23$ ). [E]
4. The cascade Coxeter rung  $D = 30 = h(E_8)$  ( $n=15$ ) ties to the exponent-centering audit  $\sum |m-15| = 56$ . [E]

**Honest separation**

The *dimension/rung coincidences* are exact arithmetic [E] — they show the old and new constructions live on one  $E_8$  spine. The old cascade's *scale assignments*  $\varphi_n$  (EW at  $n=12$ , hadronic at  $n=15, 17$ , cosmo at  $n=25$ ) are a *different* mechanism from the current  $\varphi_0^{\text{ret}}$ -power mass ladder: the cascade orders *orders of magnitude* (which sector sits where) via  $e^{-\gamma}$ , while the  $\varphi_0^{\text{ret}}$ -ladder gives *values within* a sector. Both are real, complementary orderings; neither is forced onto the other. The per-sector  $\zeta$ -calibrated absolute scales (proton mass, EW masses, CMB temperature) remain scheme-level — audited one by one in §12, with the proton mass the single explicitly open cross-sector number.

**Bottom line.** The old  $E_8$  cascade was an early, correct intuition: the physically relevant numbers are exactly the even-integer  $E_8$  orbit spine  $\{8, \dots, 60\}$ . The current work *explains why* — they are the  $(D_5 \times A_3) + \mu_4$  compiler atoms and their  $E_8$  branching blocks — and identifies the spine's two ends as the compiler boundary ( $D_{\text{start}} = 60$ ,  $\det R = 8$ ). Old and new are the same structure at different resolution.

## Part III

# The self-consistency loop

**The question, and the honest answer.** The theory rests on two inputs,  $c_3 = \frac{1}{8\pi}$  and  $g_{\text{car}} = 5$ . The old theory had a Möbius seam and the seed was the *start* of the  $E_8$  cascade. Run it *backwards*: can the cascade/ $E_8$  *return*  $c_3$  and  $g_{\text{car}}$ , closing a self-consistent loop? **Answer:** the *discrete* content is a closed, overdetermined fixed point —  $g_{\text{car}} = 5$  and the “8” in  $c_3$  are forced by the  $E_8$  closure. The *only* irreducible primitive left is the geometric  $\pi$  (the Möbius/boundary Gauss–Bonnet). So “two axioms” collapses to “*one* continuous primitive  $\pi$  + *one* discrete self-consistency fixed point.”

## 14 The loop for $g_{\text{car}}$ — an overdetermined fixed point

Two *independent* structural conditions, both pure Lie-theory, force  $g_{\text{car}} = 5$  once  $E_8$  is the closure object and  $A_3$  is the four-puncture family geometry:

### $g_{\text{car}} = 5$ is forced twice over

(A) rank fill.

The glue  $D_{g_{\text{car}}} \oplus A_3 \hookrightarrow E_8$  must fill the rank:  $\text{rank } D_{g_{\text{car}}} + \text{rank } A_3 = \text{rank } E_8$ , i.e.  $g_{\text{car}} + 3 = 8 \Rightarrow g_{\text{car}} = 5$ .

(B) Coxeter match.

The carrier Coxeter number must equal the  $E_8$  rank,  $h(D_{g_{\text{car}}}) = 2g_{\text{car}} - 2 = 8 \Rightarrow g_{\text{car}} = 5$ .

Both give  $g_{\text{car}} = 5$ , and the loop *closes*:  $g_{\text{car}} = 5 \Rightarrow D_5 \oplus A_3 \xrightarrow{\mu_4} E_8 \Rightarrow h(E_8) = 30 = 2 \cdot 3 \cdot 5 \Rightarrow \text{largest prime} = 5 = g_{\text{car}}$ . [E]

### Reverse glue selection — $D_5$ and $A_3$ from unimodular closure

The closure even *selects the two factors*, not just the rank. The discriminant of  $A_{\mu-1}$  is  $\mathbb{Z}_\mu$  and of  $D_g$  (odd  $g$ ) is  $\mathbb{Z}_4$ ; a common cyclic glue forces  $\mu = 4$ , hence  $A_{\mu-1} = A_3$ . The even-glue norm  $q(D_g) + q(A_3) = \frac{g}{4} + \frac{3}{4} = 2$  then forces  $g = 5$ , hence  $D_5$ . A second, independent confirmation of  $\mu = 4$  is the *mirror identity*  $\dim D_{\mu+1} + \dim A_{\mu-1} = \dim V_{D_{\mu+1}} \cdot |R^+(A_{\mu-1})|$ , i.e.  $3\mu(\mu+1) = \mu(\mu+1)(\mu-1) \Rightarrow \mu-1 = 3 \Rightarrow \mu = 4$  (the  $45 + 15 = 10 \cdot 6$  identity). So

$$\text{common } \mathbb{Z}_4 \text{ glue} + \text{root norm } 2 \implies D_5 + A_3$$

—  $g_{\text{car}} = 5$  is recovered from the closure, not just posited. [E]

### Familyful $E_8$ -closure fixed-point theorem

Among rank-filling pairs  $D_g \oplus A_m$  ( $g+m=8$ ) admitting an integer glue index  $\sqrt{\det D_g \det A_m} = \sqrt{4(m+1)}$ , exactly two close to an even unimodular lattice:  $D_8 \oplus A_0$  (no family) and  $D_5 \oplus A_3$  (familyful). The unique familyful solution is  $D_5 \oplus A_3$ , with lattice  $\cong E_8$  and  $g_{\text{car}} = 5$ .

| $D_g + A_m$                   | $g+m$ | glue idx <sup>2</sup> =4(m+1) | integer?                      | status                                    |
|-------------------------------|-------|-------------------------------|-------------------------------|-------------------------------------------|
| $D_8 + A_0$                   | 8     | 4                             | 2                             | valid, <i>no family</i>                   |
| $D_7 + A_1$                   | 8     | 8                             | —                             | no integer glue                           |
| $D_6 + A_2$                   | 8     | 12                            | —                             | no integer glue                           |
| <b><math>D_5 + A_3</math></b> | 8     | 16                            | <b>4= <math>\mu_4</math> </b> | <b>valid, familyful, <math>E_8</math></b> |
| $D_4 + A_4$                   | 8     | 20                            | —                             | no integer glue                           |

Equivalently, with  $q(A_{\mu-1}) = \frac{\mu-1}{\mu}$  and  $g = 9 - \mu$  the norm closure  $q(D_g) + q(A_{\mu-1}) = 2$  becomes  $\mu^2 - 5\mu + 4 = 0$ , whose nontrivial root is  $\mu = 4$  ( $A_3, g = 5$ ). So  $g_{\text{car}} = 5$  is forced *three* ways: rank-fill, Coxeter-match, and integer-glue/norm closure. [E] [verification/v6\_bootstrap.py]

**Bootstrap classification:  $D_5 \oplus A_3$  is the *unique* familyful  $E_8$  glue [E]**

$E_8$  is the unique even unimodular rank-8 lattice. Its two-factor root-lattice gluings  $L_1 \oplus L_2$  (rank = 8) that admit a *cyclic* glue are exactly those whose discriminant groups are isomorphic and cyclic:

$$D_5 \oplus A_3 (\mathbb{Z}_4), \quad E_6 \oplus A_2 (\mathbb{Z}_3), \quad E_7 \oplus A_1 (\mathbb{Z}_2), \quad A_4 \oplus A_4 (\mathbb{Z}_5).$$

Imposing the two TFPT data — one factor with a 16-dimensional half-spinor ( $\dim S_{D_g}^+ = 2^{g-1} = 16 \Leftrightarrow g = 5$ ) and one factor giving exactly  $N_{\text{fam}} = 3$  families (rank  $A_m = m = 3$ ) — selects

$$D_5 \oplus A_3 \text{ (uniquely), glue group } \mathbb{Z}_4 = |\mu_4|.$$

$E_6 \oplus A_2$  gives only 2 families,  $E_7 \oplus A_1$  only 1;  $D_8$  has non-cyclic glue  $(\mathbb{Z}_2)^2$  and  $A_8$  is a single rank-8 factor. So the carrier+family decomposition is forced. [verification/v15\_bootstrap\_classification.py]

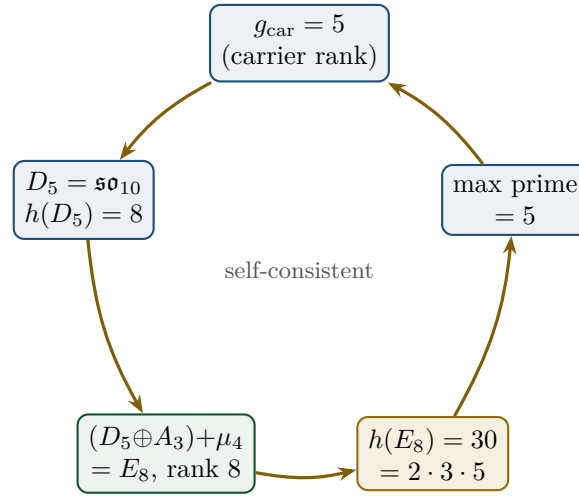


Figure 4: The closed loop:  $g_{\text{car}} = 5$  builds  $E_8$ , whose Coxeter number  $30 = 2 \cdot 3 \cdot 5$  returns the carrier prime 5. The value  $g_{\text{car}} = 5$  is the unique fixed point (overdetermined by rank-fill *and* Coxeter-match).

The carrier split  $3 + 2$  and the one-generation count  $\dim S^+ = 2^{g_{\text{car}}-1} = 16$  are then automatic, as is the glue norm  $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2$ . The family rank is rank  $A_3 = 3 = N_{\text{fam}}$  and the glue index is  $h(A_3) = 4 = |\mu_4|$  — the Coxeter numbers of the three players are  $(h(A_3), h(D_5), h(E_8)) = (4, 8, 30) = (|\mu_4|, \det R, \text{compiler})$ .

## 15 The loop for $c_3$ — the “8” is the $E_8$ rank

The boundary normalisation is  $c_3 = 1/(8\pi)$ . Its integer is exactly the carrier Coxeter number, which the loop above equals to the  $E_8$  rank:

$$c_3 = \frac{1}{h(D_5) \pi} = \frac{1}{(2g_{\text{car}} - 2) \pi} = \frac{1}{8\pi}, \quad 8 = h(D_5) = \text{rank } E_8 = \det R = \varphi(30)$$

where  $\varphi(30) = 8$  is Euler’s totient (the number of primitive residues mod  $30 =$  the  $E_8$  exponents) — so the “8” has *five* concordant readings. The tree seed  $\varphi_{\text{tree}} = 1/(6\pi)$  has  $6 = |R^+(A_3)| = h(E_8)/g_{\text{car}}$ , the three Gauss–Bonnet boundary cycles ( $6\pi = 3 \times 2\pi$ ). So the *discrete* parts of both  $c_3$  and  $\varphi_{\text{tree}}$  are  $E_8/\text{carrier}$  invariants; only the factor  $\pi$  is genuinely continuous.

**Honest caveat on  $8\pi$** 

$8\pi$  is *also* the standard Einstein normalisation ( $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ ). The loop *identifies* this gravitational 8 with rank  $E_8 = h(D_5)$  — a non-coincidence *if*  $E_8$  is the closure, but one should be clear that  $8\pi$  has an independent GR reading. The loop promotes the two 8's to the same number; it does not derive  $\pi$ .

**16 The Möbius origin of the irreducible  $\pi$** 

What the loop does *not* remove is  $\pi$ . In the old theory this is exactly the Möbius datum: the orientable double cover of the Möbius fibre is a cylinder with two boundaries plus one  $\mathbb{Z}_2$  seam  $\Gamma$ ; Gauss–Bonnet gives  $\oint k ds = 2\pi + 2\pi + 2\pi = 6\pi$  (hence  $\varphi_{\text{tree}} = 1/(6\pi)$ ), and the  $\mathbb{Z}_2$  identification is the sheet “2” of  $30 = 2 \cdot 3 \cdot 5$ . So the Möbius seam supplies precisely the two things the discrete loop cannot: the continuous  $\pi$  and the sheet parity  $\mathbb{Z}_2$ .

**Why “Möbius” is not incidental: it is the boundary Lorentz structure.** The Möbius group of the Riemann sphere is *exactly* the proper orthochronous Lorentz group:  $\text{Möb}(\hat{\mathbb{C}}) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$ , with the sphere realised as the celestial sphere of null directions via stereographic projection (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, CUP 1984, §1.2–1.4). This is the spacetime meaning of the self-consistency loop: the seam’s boundary-2-sphere carries the Lorentz symmetry, the carrier clock  $z \mapsto iz$  is an elliptic  $\text{PSL}(2, \mathbb{C})$  rotation of that sphere (the order-4 Möbius map, [verification/v180\_clock\_is\_mobius.py]), and the  $\text{RP} \rightarrow \text{OS}$ -reconstructed causal cone (the shared  $v_\gamma = v_g = c$  of `tfpt_horizon_readouts`) is the same Lorentz structure read off the seam. So “Möbius origin of  $\pi$ ” and “one Lorentzian cone” are two faces of one boundary  $\text{PSL}(2, \mathbb{C})$  — not a coincidence of names. [O] ( $\pi$  stays primitive; the identification is a structural reading).

**Net: two axioms  $\rightarrow$  one primitive + one fixed point.**

$$\{c_3, g_{\text{car}}\} \longrightarrow \underbrace{\pi \text{ (Möbius/Gauss–Bonnet)}}_{\text{one continuous primitive}} + \underbrace{E_8 \text{ closure}}_{\text{one discrete fixed point}}$$

The discrete data  $\{2, 3, 4, 5, 6, 8, 30\}$  are *all* self-consistent outputs of the  $E_8$  closure ( $g_{\text{car}} = 5$ ,  $h(D_5) = 8$ ,  $h(A_3) = 4 = |\mu_4|$ ,  $h(E_8) = 30 = 2 \cdot 3 \cdot 5$ ,  $|R^+(A_3)| = 6$ ). The only irreducible input on the *dimensionless* axis is  $\pi$ . [E] (discrete loop) / [O] ( $\pi$  is primitive)

On the *dimensionful* axis there is exactly *one* more irreducible: a single mass scale. But even that is no longer a *free* constant — fixing the physical  $\Lambda$  branch pins  $\bar{M}_{\text{Pl}}$ , so  $G_N$  is a  $\Lambda$ -metrology *output* (`origin_theory`,  $\Lambda$ -typing); the one genuinely irreducible scale is the induced-gravity (Sakharov) anchor. So the complete honest reduction is “ $\pi$  + the  $E_8$ -closure fixed point + *one* dimensionful scale”.

*One sharpening of the “8”.* Both  $\pi$ -coefficients are  $2\pi$  times a compiler integer:  $\varphi_{\text{tree}} = 1/(6\pi)$  with  $6 = 2N_{\text{fam}}$  (three Gauss–Bonnet boundary circles) and  $c_3 = 1/(8\pi)$  with  $8 = 2|\mu_4|$  (seam winding), so  $\varphi_{\text{tree}} = (|\mu_4|/N_{\text{fam}})c_3 = \frac{4}{3}c_3$ . The *same* integer 8 is the Hawking/Einstein coefficient ( $G_{\mu\nu} = 8\pi T_{\mu\nu}$ , Appendix H): the seam normaliser, the  $E_8$  rank and the universal horizon factor are one number, overdetermined ( $8 = 2|\mu_4| = \text{rank } E_8 = h(D_5) = \varphi(30) = \det R$ ). [E] [verification/v54\_seam\_horizon\_keystones.py]



### A cyclic element *inside* the hull: the order-30 Coxeter rotation [E] ([verification/v55\_coxeter\_cycle.py])

The compiler hull carries a *literal* cyclic symmetry. Built from the  $E_8$  Cartan matrix, the Coxeter element  $c = s_1 \cdots s_8$  has **order exactly**  $h(E_8) = 30 = |\mathbb{Z}_2| \cdot N_{\text{fam}} \cdot g_{\text{car}} = 2 \cdot 3 \cdot 5$  (verified by direct matrix powers), and its eigenvalues are the primitive 30th roots  $e^{2\pi i m/30}$  with  $m$  the  $E_8$  exponents  $\{1, 7, 11, 13, 17, 19, 23, 29\}$  — which are precisely the  $\varphi(30) = 8$  *totatives* of 30. Hence

$$\text{rank } E_8 = 8 = \varphi(30) = \#\{\text{exponents}\} = \#\{\text{live phases of the order-30 cycle}\}$$

So the 8 in  $c_3 = \frac{1}{8\pi}$  is the count of live phases of a primitive order-30 cyclic rotation of the hull, whose period  $30 = |\mathbb{Z}_2| \cdot N_{\text{fam}} \cdot g_{\text{car}}$  is the product of the three core integers. The cyclic structure is *present in the mathematics*, not interposed.

### The compiler atoms are the $E_8$ Casimir degrees [E] ([verification/v66\_e8\_casimir\_degrees.py])

The load-bearing integers are not an ad-hoc list: they are the fundamental *invariant degrees* of  $E_8$  (exponents +1),

$$\deg(E_8) = \{2, 8, 12, 14, 18, 20, 24, 30\},$$

with the clean (non-trivial) readings  $2=|\mathbb{Z}_2|$ ,  $8=\text{rank } E_8$ ,  $14=\dim G_2$  (the  $[K, Q]$  commutator  $G_2$ , flavor sector),  $20=\det L$ ,  $24=|W(A_3)|=4!$ ,  $30=h(E_8)$  (and the fittable  $12=|R(A_3)|$ ,  $18=p_4(a)=2+2^4$ ). Two exact sums close it:

$$\begin{aligned} \sum \deg(E_8) &= 128 = 2^7 \quad (\text{the de Sitter prefactor } 128c_3^4; 7=\text{scalaron exp}), \\ \sum \exp(E_8) &= 120 = |R^+(E_8)|. \end{aligned}$$

Since  $E_8$  is the audit hull, its invariant degrees being the theory's integers is *structurally expected* — an organizing simplification that confirms the  $E_8$  choice rather than a coincidence.

### Two-level foundation, and the parameter audit

**Local foundation:**  $c_3, g_{\text{car}}$  are the two operational inputs (the two axioms). **Global bootstrap foundation:**  $g_{\text{car}}$  and the integer in  $c_3$  are  $E_8$ -closure fixed points; only  $\pi$  is primitive. The audit:

| former input                 | status now           | recovered by                                                                 |
|------------------------------|----------------------|------------------------------------------------------------------------------|
| $g_{\text{car}} = 5$         | discrete fixed point | rank-fill, Coxeter-match, integer-glue/norm, $h(E_8)$                        |
| 8 in $c_3$                   | discrete fixed point | $h(D_5)$ , rank $E_8$ , $\det R$ , $\varphi(30)$ , $2 \mu_4 $ (seam winding) |
| 6 in $\varphi_{\text{tree}}$ | discrete fixed point | $ R^+(A_3)  = h(E_8)/g_{\text{car}}$                                         |
| $\pi$                        | geometric primitive  | Möbius / Gauss–Bonnet ( <i>not</i> derived)                                  |
| $\varphi_0^{\text{ret}}$     | derived readout      | $\frac{4}{3}c_3 + 48c_3^4$                                                   |

**Correct one-liner:** *TFPT has no freely tunable fundamental numbers left; the only remaining continuous origin is  $\pi$  from the Möbius boundary geometry.* (Not “zero axioms”.) [E]/[O]

## 17 Honest status: bootstrap, not creation from nothing

- **What is genuinely shown [E]:**  $g_{\text{car}} = 5$  is an *overdetermined* fixed point of the  $E_8$  closure (two independent Lie-theory conditions); the integer “8” in  $c_3$  is the carrier Coxeter =  $E_8$  rank; the discrete data closes a loop. The earlier landscape scan confirms  $g_{\text{car}} = 5$  is the *unique* value at which all hierarchies/family/gauge close — so the fixed point is unique, not one of many. **This uniqueness is now machine-formalised [E] in Lean 4** (Theorem A, **lean4-carrier-rigidity**): the division-free Pascal condition  $2^g = g^2 + g + 2$  has the *unique* solution  $g = 5$  (**carrier\_rank\_pascal\_unique**; growth lemma for  $g \geq 6$ ;  $N_{\text{fam}} = (2^4 - 1)/5 = 3$ ,  $g + N_{\text{fam}} = 8$ ), audited to use only the standard kernel axioms. The 2026-06-11 hardening round adds two formal modules: **AnchorLadder.lean** (the anchor power-sum law  $p_n(a) = 2 + 2^n$  for



all  $n$ , the ladder atoms (3, 4, 6, 10), 240/8/248, and the binary ladder  $p_{n+1}-p_n = 2^n$  as general theorems) and `PascalLadder.lean` (the odd-midpoint ladder identity  $\sum_{k \leq K} \binom{2K+1}{k} = 2^{2K}$  for all  $K$  via Mathlib, plus the kernel-decidable bounded uniqueness of the  $K=2$  carrier case) — the solves-side of the Pascal Ladder Theorem ([`verification/v108_pascal_ladder.py`]) is thereby [E].

- **What this is *not*:** a derivation from zero inputs. A self-consistency loop is consistent *at* a value; uniqueness is what makes it predictive (and that is supplied by the landscape scan). The continuous  $\pi$  is not produced by the loop — it is the Möbius/boundary geometry, the one irreducible primitive.
- **Conceptual gain:** the theory’s foundation tightens from “two free axioms” to “*the*  $E_8$ -closure fixed point + the geometric  $\pi$ .” That is the Möbius self-consistency you intuited: the seed is the start of the cascade, and the cascade returns the seed’s discrete content.

## Part IV

Open items and the  $E_8$ -slice scan

## 18 The honest open-items list

This is the complete current ledger of what is *not* fully closed, graded by how hard the gap is.

| #                                                                                     | Open item                                                  | Grade   | Hardening path / what closes it                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------|------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>1. SM core — finite computations, no new physics</i>                               |                                                            |         |                                                                                                                                                                                                                   |
| 1a                                                                                    | absolute quark $c$ -scale (ratios closed)                  | [O]     | quark <i>ratios</i> are integer Plücker readouts on the derived selector stratum (v49/v69/v71); only the <i>absolute</i> $U_f^*$ normalisation remains — an anchor                                                |
| 1b                                                                                    | $\theta_{12}$ solar angle                                  | [E]     | <b>texture closed (below):</b> explicit $\mu\tau$ -symmetric texture verified; only $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} \approx c_3$ (0.23%) stays conditional                                       |
| <i>2. Scheme layer — standard RG, no new input</i>                                    |                                                            |         |                                                                                                                                                                                                                   |
| 2a                                                                                    | $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$                 | [C]     | RG threshold matching $R_{\text{SM}}; m_H$ on the closed UV quartic $1/16\pi^2$                                                                                                                                   |
| 2b                                                                                    | hadron pole spectrum (singlet)                             | [C]     | singlet algebra $\mathcal{A}_{\text{sing}}$ explicit; dynamics already closed (gap $6 \log \frac{3}{2}$ )                                                                                                         |
| <i>3. Conditional — named analytic hypotheses</i>                                     |                                                            |         |                                                                                                                                                                                                                   |
| 3a                                                                                    | gravity: local eq. parameter-free; ambient redundancy      | [E]/[C] | full covariant $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ (v358/v359); perturbative graviton unitarity established (Stelle ghost a truncation artefact, v304/v370/v380); ambient measure a [C] redundancy (v369) |
| 3b                                                                                    | strong CP / QFT closure                                    | [C]     | reflection positivity (RP) is the one remaining input                                                                                                                                                             |
| 3c                                                                                    | $\Lambda$ branch label (rec vs car)                        | [C]     | one discrete choice; typed via $(8\pi)^2$ , value not fixed                                                                                                                                                       |
| <i>4. Axioms — not eliminable, only hardenable</i>                                    |                                                            |         |                                                                                                                                                                                                                   |
| 4                                                                                     | P1 ( $c_3$ ), P2 ( $g_{\text{car}}=5$ )                    | [O]     | more Lean; P2 algebra already machine-verified; remain inputs                                                                                                                                                     |
| <i>5. Frontier — typed transfer targets, never compiler powers (see frontier doc)</i> |                                                            |         |                                                                                                                                                                                                                   |
| 5                                                                                     | $\eta_B, m_p/m_e$ , Koide, dark matter, ambient QG measure | [C]     | transfers now runnable solvers with kill tests (v371–v375); ambient measure a [C] redundancy (v369); not forced (scan results below)                                                                              |
| <i>6. Cosmology late-time pipeline — not simplified by the compiler</i>               |                                                            |         |                                                                                                                                                                                                                   |
| 6                                                                                     | $C_\ell$ transfer, $a_{\ell m}$ seam, baryogenesis         | [C]     | Paper-7 machinery; $a_{\ell m}$ “good CMB world, not yet this CMB world” (SMICA test)                                                                                                                             |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in *verification/status\_ledger.csv*.

**Net.** Items 1 are finite arithmetic; 2 are standard RG; 3 hinge on named hypotheses (RP, the  $\Lambda$  branch, the UV completion); 4 are the two irreducible axioms; 5–6 are honestly open. The inflation sector (the only cosmology piece the compiler *does* sharpen) is now a parameter-free prediction ( $M=c_3^{7/2}\bar{M}_{\text{Pl}}$ , document 2).

#### Item 1b closed: the explicit solar-angle Majorana texture [E]

The last open SM angle is now realised by an *explicit* texture, not a fitted number. Take the  $\mu\tau$ -symmetric Majorana matrix (tri-bimaximal at  $\varepsilon=0$ )

$$M_\nu(\varepsilon) = \begin{pmatrix} -\eta(\varepsilon) & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad \eta(\varepsilon) = \frac{2\sqrt{2}}{\tan 2\theta_{12}} - 1, \quad \sin^2 \theta_{12} = \frac{1}{3}(1 - 2\varepsilon). \quad (1)$$

$\mu\tau$ -symmetry forces  $\theta_{23} = 45^\circ$  and  $\theta_{13} = 0$  *exactly*; diagonalisation at the TFPT solar deviation  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}}$

gives

$$\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747, \quad \theta_{23} = 45^\circ, \quad \theta_{13} = 0 \quad [\mathbf{E}] \quad (2)$$

**Honest residual** ([verification/v9\_neutrino\_texture.py]). The texture and the angle are exact. The seam identification is  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = \frac{3}{4}\left(\frac{1}{6\pi} + \frac{3}{256\pi^4}\right) = c_3 + \frac{9}{1024\pi^4}$ : the *leading geometric term is exactly*  $c_3 = \frac{1}{8\pi}$ , and the 0.23% residual is purely the  $\varphi_0^{\text{ret}}$  seed tail (document 2). So  $\varepsilon = c_3$  holds at leading order; the full  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}}$  is what enters the texture. The reactor angle  $\theta_{13}$  ( $\sin^2 \theta_{13} = e^{-5/6}\varphi_0^{\text{ret}}$ , document 2) enters through a separate  $\mu\tau$ -breaking term and is not part of this minimal solar texture. **Scope:** this realises the TFPT solar shift in the  $\mu\tau$ -symmetric *limit*; it is *not* the full PMNS matrix. The  $\theta_{23}=45^\circ$  value is the symmetric-limit centre, *not* an octant prediction — current global fits (NuFIT 6.0) leave the octant unresolved — and the reactor angle plus the combined matrix require the separately stated  $\mu\tau$ -breaking channel.

## 19 The $E_8$ -slice scan: results (honest hit / miss)

All seven maximal slices of 248 were scanned for the unassigned quantities. The rule was a meaningful (low-complexity) compiler/slice expression matching to  $< 0.5\%$ ; matches that need a free fitted factor are reported as *misses*.

| Slice / target                       | Result                                                                                                                                                                                             | Verdict                   |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| $A_4 \times A_4 = SU(5)^2$           | $\sin^2 \theta_W _{\text{GUT}} = 3/8$ sits in the $SU(5)$ hypercharge embedding; the 12 coset states are the proton-decay $X, Y$ bosons                                                            | $[\mathbf{E}]$ (known)    |
| $F_4 \times G_2$ / Koide             | Koide $Q = 2/3 =  \mathbb{Z}_2 /N_{\text{fam}}$ (sheet over families); $J_3(\mathbb{O})$ has $\dim 27=26+1$ , $3 \times 3$ octonionic “generations” on the diagonal                                | $[\mathbf{C}]$ structural |
| $D_8 = SO(16)$                       | $128 = (16, 4) \oplus (\overline{16}, \overline{4})$ — <i>same</i> spinor as $D_5 \times A_3$ ; $\nu_R$ (seesaw) lives in the 16; <b>no new dark-matter state</b> (only the 4th $\mu_4$ direction) | $[\mathbf{E}]$            |
| $A_8 = SU(9)$ , $9=3 \times 3$       | $\Lambda^3(3, 3) = (10, 1) \oplus (8, 8) \oplus (1, 10)$ : the $(8, 8)=64$ dominates — naive family $\times$ colour <b>fails</b>                                                                   | miss                      |
| $A_8 \supset SU(5) \times SU(4)$     | $\Lambda^3(5+4)$ contains the SM <b>10</b> : a genuine chiral-family-unification <i>lead</i> , but not TFPT-forced                                                                                 | lead                      |
| $E_6 \times A_2$ / Jarlskog          | cross term $2(\mathbf{1}^\top Ra)N_{\text{fam}} = 162 = 248 - 78 - 8$ exact; $J \sim \lambda_Y^6$ only to right order, no clean hit                                                                | weak                      |
| $E_8$ theta shells                   | root shells $240, 2160, 6720, \dots$ ; $31=2^{q_{\text{car}}}-1$ is <i>not</i> a clean shell ratio                                                                                                 | inconcl.                  |
| $m_p/m_e = 1836.15$                  | no clean $E_8$ /compiler expression (a QCD-confinement ratio); apparent hits need a free factor                                                                                                    | <b>miss</b>               |
| $\eta_B \approx 6.1 \times 10^{-10}$ | sits <i>between</i> $c_3^6=4 \times 10^{-9}$ and $c_3^7=1.6 \times 10^{-10}$ ; right order, no integer power                                                                                       | order only                |

Marker key:  $[\mathbf{E}]$  exact / machine-proven ·  $[\mathbf{C}]$  conditional (holds under named hypotheses) ·  $[\mathbf{O}]$  open / axiom ·  $[\mathbf{X}]$  falsifiable kill test. Per-claim fine type in verification/status\_ledger.csv.

### Genuine new readings the scan produced

- **Koide** =  $|\mathbb{Z}_2|/N_{\text{fam}}$ . The empirical Koide value  $2/3$  is exactly the sheet over the family count — a clean (if simple) compiler reading, promoting Koide from “near  $2/3$ ” to “democratic  $|\mathbb{Z}_2|/N_{\text{fam}}$ .”
- **$E_8$  as the optimal 8-bit code.**  $E_8$  is the optimal 8D lattice (kissing number 240). In byte framing  $256-248 = 8 = \text{rank } E_8$  and  $256-240 = 16 = \dim S^+$  — a clean information-theoretic reading of the two load-bearing  $E_8$  numbers.
- **Dark matter = no new state.** The  $SO(16)$  scan shows the 128 is the *same* matter spinor; there is no spare  $E_8$  singlet to be a WIMP. Any DM candidate must be the determinant-line axion (frontier doc), not a new  $E_8$  rep — a useful *negative*.
- $SU(9) \supset SU(5) \times SU(4)$  **chiral lead.**  $\Lambda^3$  puts the SM **10** in the antisymmetric; the cleanest unexploited route to a family-unification reading, flagged for future work.

**Honest negatives**

$m_p/m_e$  has *no* clean  $E_8$ /compiler derivation (it is a confinement ratio — the apparent “hits” all need a free fitted factor);  $\eta_B$  matches only at the order-of-magnitude level ( $\sim c_3^{6.x}$ ); the  $\Lambda$  rec/car branch is *not* selected by the theta series; the Jarlskog invariant is right only to order  $\lambda_Y^6$ . These are reported as open, not forced.

**20 Following the  $SU(9)$  lead: cross-slice consistency**

The one scan lead worth following is  $A_8 = SU(9) \supset SU(5) \times SU(4)$ . Splitting  $9 = (5, 1) \oplus (1, 4)$  the  $E_8$  branching  $248 = 80 \oplus 84 \oplus \overline{84}$  decomposes *exactly* as:

| piece              | $SU(5) \times SU(4)$ content                                                                 | reading                                               |
|--------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------|
| 80 (adj)           | $(24, 1) + (1, 15) + (5, \overline{4}) + (\overline{5}, 4) + (1, 1)$                         | $SU(5)_{\text{GUT}} + SU(4)_{\text{fam}}$ gauge       |
| $84 = \Lambda^3 9$ | $(10, 1) + (10, 4) + (5, 6) + (1, 4)$                                                        | the SM <b>10</b> as family-singlet + family- <b>4</b> |
| $\overline{84}$    | $(\overline{10}, 1) + (\overline{10}, \overline{4}) + (\overline{5}, 6) + (1, \overline{4})$ | the conjugate matter                                  |

(Dimension checks:  $24+15+20+20+1 = 80$ ;  $10+40+30+4 = 84$ ; total 248.)

**The genuine content:  $A_8$  and  $D_5 \times A_3$  agree on the family [E]**

The  $SU(4)$  here is *the same*  $A_3$  family group as in the construction slice  $D_5 \times A_3$  — the four punctures  $\mu_4$  of  $\mathbb{P}^1 \setminus \mu_4$ . The  $SU(5)$  is the Georgi–Glashow GUT sitting inside the carrier  $SO(10) = D_5$  ( $SU(5) \subset SO(10)$ ). So the two maximal slices are *consistent*: both read the family as  $SU(4)=A_3$ , and  $A_8$  additionally exposes the  $SU(5)$  matter content — the SM **10** appears as  $(10, 1) \oplus (10, 4)$ , i.e. a family-singlet **10** plus a family-**4**, where the **4** of  $A_3$  is exactly TFPT’s “ $4 = N_{\text{fam}} + 1$ ” (three families + the  $\mu_4$  direction).

**Honest limit: chirality is still the index, not  $E_8$** 

$84 \oplus \overline{84}$  is *vectorlike* ( $10 \oplus \overline{10}$  etc.):  $E_8$  being non-chiral, the slice cannot by itself produce three *chiral* generations. The chirality is the boundary index  $\text{Ind}(D_{\text{fam}}^+) = 3$  — the *same* input TFPT already uses. So  $SU(9)$  is a consistent re-reading (matter in  $SU(5)$ , family in  $A_3$ ), not an independent derivation of  $N_{\text{fam}} = 3$ . Each  $SU(5)$  pair (**10** + **5**) is anomaly-free, so the embedding is clean. [C]

**Appendix: computational verification**

The audit/bridge identities of this document (marked [E], [E]) are re-derived from  $\{c_3, g_{\text{car}}\}$  and machine-checked by the suite in the repository folder `verification/`; the inline tags `[verification/...]` point to the exact script. Relevant here:

- `v1_e8_glue.py` — the  $\mu_4$  glue  $E_8 = (D_5 \oplus A_3) + \mu_4$  (load-bearing).
- `v4_flavor_matrix.py` — the spectral selector  $\det R = h(D_5) = 8$ , minors  $\{2, 3, 5\}$ .
- `v5_e8_cascade.py` — the cascade  $D_n = 60 - 2n$ : endpoints, exponent rungs, IR tail, variance.
- `v6_bootstrap.py` — the self-consistency loop:  $\mu^2 - 5\mu + 4 = 0$ ,  $g_{\text{car}} = 5$  three ways,  $8 = \text{rank } E_8$ .

The  $E_8$ -slice/atlas readings are explicitly [O] (audit-level), not part of the pass/fail suite. Run `python verification/run_all.py`; full map in `verification/README.md`.